博客园 cnblogs.com 会员 周边 捐助 新闻 博问 闪存 赞助商 代码改变世界 注册 登录

收录查询

多线程脑袋

博客园 首页 联系 订阅 管理

公告

昵称: ->
园龄: 20年1个月
粉丝: 11
关注: 0
+加关注

< 2024年10月 >

日	一	二	三	四	五	六
29	30	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2
3	4	5	6	7	8	9

搜索

找找看

常用链接

我的随笔
我的评论
我的参与
最新评论
我的标签

积分与排名

积分 - 115427
排名 - 12796

随笔分类 (19)

MSDN文章or链接(1)
TEMP代码垃圾桶(4)
大家来砸场子啊(2)
文章集[非技术类](4)
文章集[技术类](8)

文章分类 (419)

CP等站点地图(1)
CP网站名人堂(8)
Foo C#(26)
Foo C++(49)
Foo DataBase(1)
Foo DataStructure(6)
Foo Debuger(17)
Foo Encryption(4)
Foo File Format(12)
Foo FlashMX(3)
Foo Game(2)
Foo HardWare(3)
Foo HTML&&CSS(4)
Foo JAVA(5)
Foo JavaScript(23)

随笔 - 25 文章 - 422 评论 - 29 阅读 - 26万

crt0.c代码内容解释和编译器构造

下面用红色字体，做部分解释：

```
/***
*crt0.c - C runtime initialization routine
*
*       Copyright (c) 1989-1997, Microsoft Corporation. All rights reserved.
*
*Purpose:
*       This the actual startup routine for apps.  It calls the user's main
*       routine [w]main() or [w]WinMain after performing C Run-Time Library
*       initialization.
*
*       (With ifdef's, this source file also provides the source code for
*       wcrt0.c, the startup routine for console apps with wide characters,
*       wincrt0.c, the startup routine for Windows apps, and wwincrt0.c,
*       the startup routine for Windows apps with wide characters.)
*
*******************************************************************/

#ifdef _WIN32

#ifndef CRTDLL

#include <cruntime.h>
#include <dos.h>
#include <internal.h>
#include <stdlib.h>
#include <string.h>
#include <rterr.h>
#include <windows.h>
#include <awint.h>
#include <tchar.h>
#include <dbgint.h>

/*
 * wWinMain is not yet defined in winbase.h. When it is, this should be
 * removed.
 */

int
WINAPI
wWinMain(
    HINSTANCE hInstance,
    HINSTANCE hPrevInstance,
    LPWSTR lpCmdLine,
    int nShowCmd
    );

```

Foo Linux/UNIX(61)
Foo MASM(3)
Foo MFC(14)
Foo OS(12)
Foo PHP(1)
Foo protocol(10)
Foo RFC(2)
Foo Schema(1)
Foo Shell(16)
Foo SQLServer(13)
Foo UI Design(3)
Foo VC Optimize(1)
Foo virus(1)
Foo WIN32SDK(16)
Foo XML(10)
Foo编译原理(14)
MSDN[Archi](12)
MSDN2001[AC](9)
TEMP文章[粘了再说](6)
VStudio Family(11)
Windows维护管理(10)
非技术文章(10)
名人堂(2)
时间管理(7)
英语学习(10)

[国外]纯WIN32编程
Data Compression/Decompression
TOP4
TOP3
TOP2
TOP1
MVP Tips
TOP6
TOP5
CS Courses
The C Library Reference Guide

English OL Dictionary
短语
phrase
图文词典[棒]
数理化词典
不规则动词词典
Irregular Verb Dictionary
在线百科全书
encyclopedia
格言词典2
Quotation
传记词典
Biographies
俚语词典
SLang
格言词典
Quotes
动词词典
Phrasal Verb Dictionary

```
#ifdef WPRFLAG
_TUCHAR * __cdecl _wwincmdln(void);
#else  /* WPRFLAG */
_TUCHAR * __cdecl _wincmdln(void);
#endif  /* WPRFLAG */

/*
 * command line, environment, and a few other globals
 */

#ifdef WPRFLAG
wchar_t *_wcmdln;           /* points to wide command line */
#else  /* WPRFLAG */
char *_acmdln;              /* points to command line */
#endif  /* WPRFLAG */

char *_aenvptr = NULL;      /* points to environment block */
wchar_t *_wenvptr = NULL;   /* points to wide environment block */


void (__cdecl * _aexit_rtn)(int) = _exit;   /* RT message return procedure */

static void __cdecl fast_error_exit(int);   /* Error exit via ExitProcess */

/*
 * _error_mode and _apptype, together, determine how error messages are
 * written out.
 */
int __error_mode = _OUT_TO_DEFAULT;
#ifdef _WINMAIN_
int __app_type = _GUI_APP;
#else  /* _WINMAIN_ */
int __app_type = _CONSOLE_APP;
#endif  /* _WINMAIN_ */


/***
*BaseProcessStartup(PVOID Peb)
*
*Purpose:
*       This routine does the C runtime initialization, calls main(), and
*       then exits.  It never returns.
*
*Entry:
*       PVOID Peb - pointer to Win32 Process Environment Block (not used)
*
*Exit:
*       This function never returns.
*
*******************************************************************/

#ifdef _WINMAIN_

#ifdef WPRFLAG
void wWinMainCRTStartup(
#else  /* WPRFLAG */
void WinMainCRTStartup(
#endif  /* WPRFLAG */
```

Eook of science
EBookMall

PhotoShop Guru
GUI锦标赛
CG橙子
FPSD1
Foreign PhotoShop Designer

PHP资源集
IBM的PHP资源中心

UNIX 和 Linux
IBM网站Linux专栏
UNIX ported tools
电子书Online:Securing and Optimizing Linux
GCC的stdc++
GCC文档Arch
GCC总站
英文LinuxFocus
Linux Focus
GNU手册页
Red Hat 9在线手册(中文)
鸟哥的Linux私房菜
Linux At IBM
Core JavaScript 1.5 Reference
NASM手册
Linux&&UNIX Realease Doc
IBM developerWorks Linux 专栏
OS相关
GNU开发资源
Linux Files and Command Reference
GNU Make中文手册
Program Library HOWTO

凤凰电视台
偶的另一片地盘
李敖有话说

国内技术站点
百度C论坛
VC知识库BBS
Foo Game Design
西北工业大学CCS
数据结构和算法（基础）
UML和软件工程
吕震宇的BLOG
邪恶八进制
CSDN唯一有价值的也就只有它了
C++FAQ中文站
中国STL学习中心
测试时代
VCKBASE的MSDN MTT首页
博客园牛人
VCKBASE常用算法列表
Foo JavaScriptEditor
Foo DIV+CSS
算法BLOG

```
#else  /* _WINMAIN_ */

#ifdef WPRFLAG
void wmainCRTStartup(
#else  /* WPRFLAG */
void mainCRTStartup(
#endif  /* WPRFLAG */

#endif  /* _WINMAIN_ */
        void
        )

{
        int mainret;

#ifdef _WINMAIN_
        _TUCHAR *lpszCommandLine;
        STARTUPINFO StartupInfo;
#endif  /* _WINMAIN_ */

        /*
         * Get the full Win32 version
         */
        _osver = GetVersion();

        _winminor = (_osver >> 8) & 0x00FF ;
        _winmajor = _osver & 0x00FF ;
        _winver = (_winmajor << 8) + _winminor;
        _osver = (_osver >> 16) & 0x00FFFF ;

#ifdef _MT
        if ( !_heap_init(1) )           /* initialize heap */
#else  /* _MT */
        if ( !_heap_init(0) )           /* initialize heap */
#endif  /* _MT */
            fast_error_exit(_RT_HEAPINIT);  /* write message and die */

#ifdef _MT
        if( !_mtinit() )                /* initialize multi-thread */
            fast_error_exit(_RT_THREAD);    /* write message and die */
#endif  /* _MT */

        /*
         * Guard the remainder of the initialization code and the call
         * to user's main, or WinMain, function in a __try/__except
         * statement.
         */

        __try {

            _ioinit();                  /* initialize lowio */

#ifdef WPRFLAG
            /* get wide cmd line info */
            _wcmdln = (wchar_t *)__crtGetCommandLineW();

            /* get wide environ info */
            _wenvptr = (wchar_t *)__crtGetEnvironmentStringsW();
```

音扑（ASP.NET论坛---速度快）
南大做的OS教学系统

国外技术站点
Doom9
system internal
DnZone
M$公司顶级Debug大师BLOG
Channel9视频
W3C标准网站
XML学习站点
Flash动作脚本教程
whois查询
wincommand
CUJM
设计模式
俄罗斯的某个MIT站点
HowStuffWorks[它怎么工作的]
[英文]维基百科
10分钟系列和STL/C++等
STL程序员指南
C++
Google的C++组
Microsoft Visual C++
STL User's Guild
查协议的RFC编号
关于RFC
RFC编号查询
RFC文档下载
（英文）RELISOFT首页
CodeProject站点
Intel中文站点
VS.NET Express站点
ASP.NET
command
http://www.winpcap.org/misc/links.htm#tools
Foo Color2
Foo Color
Foo JavaScript
AJAX应用
综合(C++/MFC/WIN32...)

图片设计
Foo Digital Photograph tutorials
中国艺术设计联盟
Foo GUITaobao
Foo Photo
某人的GUI博客
某人的BLOG

网络安全
VBRad
安全焦点TOOLS

微软Microsoft技术
MSDN DevCenter
EVENT AND WEBCASTS
MVP FAQ

```

        _wsetargv();
        _wsetenvp();
#else  /* WPRFLAG */
        /* get cmd line info */
        _acmdln = (char *)GetCommandLineA();

        /* get environ info */
        _aenvptr = (char *)__crtGetEnvironmentStringsA();

        _setargv();
        _setenvp();
#endif  /* WPRFLAG */

        _cinit();                   /* do C data initialize */

#ifdef _WINMAIN_

        StartupInfo.dwFlags = 0;
        GetStartupInfo( &StartupInfo );

#ifdef WPRFLAG
        lpszCommandLine = _wwincmdln();
        mainret = wWinMain(
#else  /* WPRFLAG */
        lpszCommandLine = _wincmdln();
        mainret = WinMain(
#endif  /* WPRFLAG */
                        GetModuleHandleA(NULL),
                        NULL,
                        lpszCommandLine,
                        StartupInfo.dwFlags & STARTF_USESHOWWINDOW
                            ? StartupInfo.wShowWindow
                            : SW_SHOWDEFAULT
                      );
#else  /* _WINMAIN_ */

#ifdef WPRFLAG
        __winitenv = _wenviron;
        mainret = wmain(__argc, __wargv, _wenviron);
#else  /* WPRFLAG */
        __initenv = _environ;
        mainret = main(__argc, __argv, _environ);
#endif  /* WPRFLAG */

#endif  /* _WINMAIN_ */
        exit(mainret);
    }
    __except ( _XcptFilter(GetExceptionCode(), GetExceptionInformation
()) )
    {
        /*
         * Should never reach here
         */
        _exit( GetExceptionCode() );

    } /* end of try - except */

}
```

The Old New Thing
《威胁和对策》
[M$]System Error Codes
Exception Handing（CP网）
（英文）MSDN站VC首页
MSDN中文站"控件"介绍
MSDN中文站VC++首页
MSDN中文站点
M$ Systems Journal 主页
MSDN杂志
MS全部产品和资料中心
AJAX
MFC GUI Controls
ASP.NET快速上手
MS.NET FrameWork参考
ASP.NET语法参考

资源和其它
Think In系列电子书下载
计算机语言发展历史
developerWorks多媒体课堂
计算机发展历史
英文电子书下载网
疯狂广告网
MIT:数学课程系列
创作共享，天下为公（OOPS）
Operating System
FIF小组网页制作

阅读排行榜
1. [总结]软件工程师笔试题目（C++）(7042)
2. The quick brown fox jumps over the lazy dog.(2064)
3. [转贴]The Code Project Visual C++ Forum FAQ(1963)
4. 看看你是否需要更新SYMBOL文件了??(1411)
5. 为什么用户自定义消息通常+0x400(1223)

推荐排行榜
1. [总结]软件工程师笔试题目（C++）(1)

最新评论
1. Re:[转]War3(魔兽争霸III)地图全开的源代码
我前段时间试过了
--->
2. Re:[转]War3(魔兽争霸III)地图全开的源代码
是可以使用的.不错o
--xiaomi123
3. re: [转]原码、反码、补码
对于补码、原码的互换，我更支持于符位号一起取反加1，而不是两者分开处理。例如，楼主的：已知一个补码为11111001，则原码是10000111（-7）：因为符号位为"1"，表示是一个负数，所以该位不变...

```c


/***
*_amsg_exit(rterrnum) - Fast exit fatal errors
*
*Purpose:
*       Exit the program with error code of 255 and appropriate error
*       message.
*
*Entry:
*       int rterrnum - error message number (amsg_exit only).
*
*Exit:
*       Calls exit() (for integer divide-by-0) or _exit() indirectly
*       through _aexit_rtn [amsg_exit].
*       For multi-thread: calls _exit() function
*
*Exceptions:
*
*******************************************************************/

void __cdecl _amsg_exit (
        int rterrnum
        )
{
#ifdef _WINMAIN_
        if ( __error_mode == _OUT_TO_STDERR )
#else  /* _WINMAIN_ */
        if ( __error_mode != _OUT_TO_MSGBOX )
#endif  /* _WINMAIN_ */
            _FF_MSGBANNER();    /* write run-time error banner */

        _NMSG_WRITE(rterrnum);  /* write message */
        _aexit_rtn(255);        /* normally _exit(255) */
}

/***
*fast_error_exit(rterrnum) - Faster exit fatal errors
*
*Purpose:
*       Exit the process with error code of 255 and appropriate error
*       message.
*
*Entry:
*       int rterrnum - error message number (amsg_exit only).
*
*Exit:
*       Calls ExitProcess.
*
*Exceptions:
*
*******************************************************************/

static void __cdecl fast_error_exit (
        int rterrnum
        )
```

--变

4. re: GCC的命令行options

我怎么都看不了啊?

--tianyongwei

5. re: [转]原码、反码、补码

很不错的基础知识文章。谢了

--RogerWang

6. re: Linux配置文件内容(3) /etc/profile[未登录]

@方法 你找一些shell编程的书看一下，如果你有C/C++基础，基本就能看懂了 再就是要熟悉LINUX中很多小的应用程序 管道和重定向是UNIX-LIKE操作系统中最重要的特性...

--shanzy

7. re: Linux配置文件内容(3) /etc/profile

能解释下么?

--方法

8. re: [转]自已动手制作linux下的拼音五笔输入法

顶了 !不错不错啊！ ： ： ...

--大法官

9. re: [转帖]李阳疯狂英语励志名言

真不错，收藏了！thank you!

--游客

10. re: [转]LoadLibrary()和 _declspec(dllexport)

调用约定啊

--刘

```
{
#ifdef _WINMAIN_
        if ( __error_mode == _OUT_TO_STDERR )
#else  /* _WINMAIN_ */
        if ( __error_mode != _OUT_TO_MSGBOX )
#endif  /* _WINMAIN_ */
            _FF_MSGBANNER();    /* write run-time error banner */

        _NMSG_WRITE(rterrnum);  /* write message */
        ExitProcess(255);       /* normally _exit(255) */
}

#ifndef WPRFLAG


#endif  /* WPRFLAG */

#endif  /* CRTDLL */

#else  /* _WIN32 */

#include <cruntime.h>
#include <internal.h>
#include <stdlib.h>
#include <msdos.h>
#include <string.h>
#include <setjmp.h>
#include <dbgint.h>
#include <macos\types.h>
#include <macos\segload.h>
#include <macos\gestalte.h>
#include <macos\osutils.h>
#include <macos\traps.h>
#include <mpw.h>

static void __cdecl Inherit(void);  /* local function */

int __cdecl main(int, char **, char **);            /*generated by compiler*/

unsigned long _GetShellStack(void);

static char * __cdecl _p2cstr_internal ( unsigned char * str );

extern MPWBLOCK * _pMPWBlock;
extern int __argc;
extern char **__argv;

/***
*__crt0()
*
*Purpose:
*       This routine does the C runtime initialization, calls main(), and
*       then exits.  It never returns.
*
*Entry:
*
*Exit:
*       This function never returns.
*
```

```
*************************************************************
******************/

void __cdecl __crt0 (
        )
{
        int mainret;
        char szPgmName[32];
        char *pArg;
        char *argv[2];

#ifndef _M_MPPC
        void *pv;

        /* This is the magic stuff that MPW tools do to get info from MPW*/

        pv = (void *)*(int *)0x316;
        if (pv != NULL && !((int)pv & 1) && *(int *)pv == 'MPGM') {
            pv = (void *)*++(int *)pv;
            if (pv != NULL && *(short *)pv == 'SH') {
                _pMPWBlock = (MPWBLOCK *)pv;
            }
        }

#endif  /* _M_MPPC */

        _environ = NULL;
        if (_pMPWBlock == NULL) {
            __argc = 1;
            memcpy(szPgmName, (char *)0x910, sizeof(szPgmName));
            pArg = _p2cstr_internal(szPgmName);
            argv[0] = pArg;
            argv[1] = NULL;
            __argv = argv;

#ifndef _M_MPPC
            _shellStack = 0;                    /* force ExitToShell */
#endif  /* _M_MPPC */
        }
#ifndef _M_MPPC
        else {
            _shellStack = _GetShellStack();       //return current a6, or first a6
            _shellStack += 4;                   //a6 + 4 is the stack pointer we wa
nt
            __argc = _pMPWBlock->argc;
            __argv = _pMPWBlock->argv;

            Inherit();      /* Inherit file handles - env is set up by _envinit if nee
ded */
        }
#endif  /* _M_MPPC */

        /*
         * call run time initializer
         */
        __cinit();

        mainret = main(__argc, __argv, _environ);
        exit(mainret);
```

```
}


#ifndef _M_MPPC
/***
*Inherit() - obtain and process info on inherited file handles.
*
*Purpose:
*
*       Locates and interprets MPW std files.  For files we just save the
*       file handles.   For the console we save the device table address so
*       we can do console I/O.  In the latter case, FDEV is set in the _osfile
*       array.
*
*Entry:
*       Address of MPW param table
*
*Exit:
*       No return value.
*
*Exceptions:
*
*******************************************************************/

static void __cdecl Inherit (
        void
        )
{
        MPWFILE *pFile;
        int i;
        pFile = _pMPWBlock->pFile;
        if (pFile == NULL) {
            return;
        }
        for (i = 0; i < 3; i++) {
            switch ((pFile->pDevice)->name) {
                case 'ECON':
                    _osfile[i] |= FDEV | FOPEN;
                    _osfhnd[i] = (int)pFile;
                    break;

                case 'FSYS':
                    _osfile[i] |= FOPEN;
                    _osfhnd[i] = (*(pFile->ppFInfo))->ioRefNum;
                    break;
            }
            pFile++;
        }
}

#endif  /* _M_MPPC */



static char * __cdecl _p2cstr_internal (
        unsigned char * str
        )
{
```

```
    unsigned char *pchSrc;
    unsigned char *pchDst;
    int  cch;

    if ( str && *str ) {
        pchDst = str;
        pchSrc = str + 1;

        for ( cch=*pchDst; cch; --cch ) {
            *pchDst++ = *pchSrc++;
        }

        *pchDst = '\0';
    }

    return( str );
}

#endif  /* _WIN32 */

```

1. 应用程序类型--------只有2类：要么是CUI， 要么是GUI

2. 程序真正的启动代码（入口点函数）----------各种CRTStartUp**

3. 环境变量+命令行信息的相关变量和函数都在此文件中有定义或声明

4. 调用运行时初始化+main()函数的最后返回和报错信息在下面的代码定义了

```
****************************************************************
****************/

void __cdecl __crt0 (
        )
{
        int mainret;
        char szPgmName[32];
        char *pArg;
        char *argv[2];

#ifndef _M_MPPC
        void *pv;

        /* This is the magic stuff that MPW tools do to get info from MPW*/

        pv = (void *)*(int *)0x316;
        if (pv != NULL && !((int)pv & 1) && *(int *)pv == 'MPGM') {
            pv = (void *)*++(int *)pv;
            if (pv != NULL && *(short *)pv == 'SH') {
                _pMPWBlock = (MPWBLOCK *)pv;
            }
        }

#endif  /* _M_MPPC */

        _environ = NULL;
        if (_pMPWBlock == NULL) {
            __argc = 1;
            memcpy(szPgmName, (char *)0x910, sizeof(szPgmName));
            pArg = _p2cstr_internal(szPgmName);
            argv[0] = pArg;
            argv[1] = NULL;
            __argv = argv;
```

```

#ifndef _M_MPPC
        _shellStack = 0;                    /* force ExitToShell */
#endif  /* _M_MPPC */
    }
#ifndef _M_MPPC
    else {
        _shellStack = _GetShellStack();       //return current a6, or first a6
        _shellStack += 4;                   //a6 + 4 is the stack pointer we want
        __argc = _pMPWBlock->argc;
        __argv = _pMPWBlock->argv;

        Inherit();      /* Inherit file handles - env is set up by _envinit if needed */
    }
#endif  /* _M_MPPC */

    /*
     * call run time initializer
     */
    __cinit();

    mainret = main(__argc, __argv, _environ);
    exit(mainret);
}


```

5. 获取进程信息+响应中断的代码如下:

```
/***
*Inherit() - obtain and process info on inherited file handles.
*
*Purpose:
*
*       Locates and interprets MPW std files.  For files we just save the
*       file handles.   For the console we save the device table address so
*       we can do console I/O.  In the latter case, FDEV is set in the _osfile
*       array.
*
*Entry:
*       Address of MPW param table
*
*Exit:
*       No return value.
*
*Exceptions:
*
*******************************************************************************/

static void __cdecl Inherit (
        void
        )
{
        MPWFILE *pFile;
        int i;
        pFile = _pMPWBlock->pFile;
        if (pFile == NULL) {
```

```
        return;
    }
    for (i = 0; i < 3; i++) {
        switch ((pFile->pDevice)->name) {
            case 'ECON':
                _osfile[i] |= FDEV | FOPEN;
                _osfhnd[i] = (int)pFile;
                break;

            case 'FSYS':
                _osfile[i] |= FOPEN;
                _osfhnd[i] = (*(pFile->ppFInfo))->ioRefNum;
                break;
        }
        pFile++;
    }
}

#endif  /* _M_MPPC */



static char * __cdecl _p2cstr_internal (
        unsigned char * str
        )
{
        unsigned char *pchSrc;
        unsigned char *pchDst;
        int  cch;

        if ( str && *str ) {
            pchDst = str;
            pchSrc = str + 1;

            for ( cch=*pchDst; cch; --cch ) {
                *pchDst++ = *pchSrc++;
            }

            *pchDst = '\0';
        }

        return( str );
}

#endif  /* _WIN32 */
```

6. 如果是GUI程序，则要创建"内核对象"，请参考下面的MSDN中的STARTUPINFO结构体的说明

http://www.cnblogs.com/shanzy/articles/513455.html

7. 上面的crt0.c文件中涉及到MS针对不同的CPU厂商都能够跑MS的VC编译器的CPU假设，如下面MSDN文章所说----------当然，你也可以在代码中手写下面这些MS预定义宏，或者写MAKEFILE

Predefined Macros

The compiler recognizes six predefined ANSI C macros (see Table 1.1), and the Microsoft C++ implementation provides several more (see Table 1.2). These macros take no arguments and cannot be redefined. Their value

(except for **__LINE__** and **__FILE__**) must be constant throughout compilation. Some of the predefined macros listed below are defined with multiple values. Their values can be set by selecting the corresponding menu option in the Visual C++ development environment, or by using a command-line switch. See the tables below for more information.

Table 1.1 ANSI Predefined Macros

Macro	Description
__DATE__	The compilation date of the current source file. The date is a string literal of the form *Mmm dd yyyy*. The month name *Mmm* is the same as for dates generated by the library function **asctime** declared in TIME.H.
__FILE__	The name of the current source file. **__FILE__** expands to a string surrounded by double quotation marks.
__LINE__	The line number in the current source file. The line number is a decimal integer constant. It can be altered with a **#line** directive.
__STDC__	Indicates full conformance with the ANSI C standard. Defined as the integer constant 1 only if the /Za compiler option is given and you are not compiling C++ code; otherwise is undefined.
__TIME__	The most recent compilation time of the current source file. The time is a string literal of the form *hh:mm:ss*.
__TIMESTAMP__	The date and time of the last modification of the current source file, expressed as a string literal in the form *Ddd Mmm Date hh:mm:ss yyyy*, where *Ddd* is the abbreviated day of the week and *Date* is an integer from 1 to 31.

Table 1.2 Microsoft-Specific Predefined Macros

Macro	Description
_CHAR_UNSIGNED	Default **char** type is unsigned. Defined when /J is specified.
__cplusplus	Defined for C++ programs only.
_CPPRTTI	Defined for code compiled with /GR (Enable Run-Time Type Information).
_CPPUNWIND	Defined for code compiled with /GX (Enable Exception Handling).
_DLL	Defined when /MD or /MDd (Multithread DLL) is specified.

_M_ALPHA	Defined for DEC ALPHA platforms. It is defined as 1 by the ALPHA compiler, and it is not defined if another compiler is used.
_M_IX86	Defined for x86 processors. See Table 1.3 for more details.
_M_MPPC	Defined for Power Macintosh platforms. Default is 601 (/QP601). See Table 1.4 for more details.
_M_MRX000	Defined for MIPS platforms. Default is 4000 (/QMR4000). See Table 1.5 for more details.
_M_PPC	Defined for PowerPC platforms. Default is 604 (/QP604). See Table 1.6 for more details.
_MFC_VER	Defines the MFC version. Defined as 0x0421 for Microsoft Foundation Class Library 4.21. Always defined.
_MSC_EXTENSIONS	This macro is defined when compiling with the /Ze compiler option (the default). Its value, when defined, is 1.
_MSC_VER	Defines the compiler version. Defined as 1200 for Microsoft Visual C++ 6.0. Always defined.
_MT	Defined when /MD or /MDd (Multithreaded DLL) or /MT or /MTd (Multithreaded) is specified.
_WIN32	Defined for applications for Win32®. Always defined.

As shown in following tables, the compiler generates a value for the preprocessor identifiers that reflect the processor option specified.

Table 1.3 Values for _M_IX86

Option in Developer Studio	Command-Line Option	Resulting Value
Blend	/GB	**_M_IX86 = 500** (Default. Future compilers will emit a different value to reflect the dominant processor.)
Pentium	/G5	**_M_IX86 = 500**
Pentium Pro	/G6	**_M_IX86 = 600**
80386	/G3	**_M_IX86 = 300**
80486	/G4	**_M_IX86 = 400**

Table 1.4 Values for _M_MPPC

Option in development environment	Command-Line Option	Resulting Value
PowerPC 601	/QP601	**_M_MPPC = 601** (Default)
PowerPC 603	/QP603	**_M_MPPC = 603**
PowerPC 604	/QP604	**_M_MPPC = 604**
PowerPC 620	/QP620	**_M_MPPC = 620**

Table 1.5 Values for _M_MRX000

Option in Developer Studio	Command-Line Option	Resulting Value
R4000	/QMR4000	**_M_MRX000 = 4000** (Default)
R4100	/QMR4100	**_M_MRX000 = 4100**
R4200	/QMR4200	**_M_MRX000 = 4200**
R4400	/QMR4400	**_M_MRX000 = 4400**
R4600	/QMR4600	**_M_MRX000 = 4600**
R10000	/QMR10000	**_M_MRX000 = 10000**

Table 1.6 Values for _M_PPC

Option in Developer Studio	Command-Line Option	Resulting Value
PowerPC 601	/QP601	**_M_PPC = 601**
PowerPC 603	/QP603	**_M_PPC = 603**
PowerPC 604	/QP604	**_M_PPC = 604** (Default)
PowerPC 620	/QP620	**_M_PPC = 620**

8. 任何C/C++编译器的构造，都是依托于ANSI C/ANSI C++标准的，但是，由于某些历史问题当ANSI标准影响了厂商的标准，或者厂商认为某些东西更利于开发的时候，厂商会在ANSI标准提供的编译器的基础上对CRT运行时库进行扩充，扩充的结果就是上面的情况

9. 厂商扩充了ANSI标准，就需要写出一个自己的编译器(MS 的VC编译器就是cl.exe)，写这个cl.exe的过程要根据厂商扩充之后的各种标准表格，在写这个cl.exe的时候做出很多的选择分支(当然，MS这里用它认为重要的一些"自定义宏"来写它的cl.exe)

分类: Foo C++

好文要顶 关注我 收藏该文 微信分享

-> 粉丝 - 11 关注 - 0 0 0
+加关注

升级成为会员

posted @ 2006-09-24 17:45 -> 阅读(2284) 评论(0) 编辑 收藏 举报

刷新页面 返回顶部

登录后才能查看或发表评论，立即 登录 或者 逛逛 博客园首页

【推荐】100%开源！大型工业跨平台软件C++源码提供，建模，组态！
【推荐】轻量又高性能的 SSH 工具 IShell：AI 加持，快人一步
【推荐】2024阿里云超值优品季，精心为您准备的上云首选必备产品

编辑推荐：
· .NET云原生应用实践（二）：Sticker微服务RESTful API的实现
· SaaS架构：开放平台架构设计
· 一张图带你了解.NET终结(Finalize)流程
· 漫谈高并发业务的 CAS 及 ABA
· 再谈如何优雅修改代码

阅读排行：
· .NET 8 实现无实体库表 API 部署服务
· 致阿里云：我有一个小需求，请帮忙去掉AI助手
· OpenAI官方开源多智能体框架「Swarm」，并不是我想要的多智能体框架
· 推荐一款支持Vue3的管理系统模版：Vue-Vben-Admin
· SaaS架构：开放平台架构设计